

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

REAL TIME PROJECT/FIELD BASED PROJECT REVIEW-STAGE-II

Temperature Control Fan Using Arduino And Temperature Sensor

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ABSTRACT

A Temperature Controlled Fan is an automatic cooling system designed to regulate the speed of a fan based on the surrounding temperature. The main aim of this project is to reduce manual effort and improve energy efficiency by automatically adjusting the fan speed according to environmental conditions. In conventional systems, fan speed is controlled manually, which can lead to unnecessary power consumption and inconvenience. In this project, a temperature sensor such as DHT 11 is used to detect the ambient temperature. The sensor converts the temperature into an electrical signal which is sent to a control unit like a microcontroller. The controller processes the signal and adjusts the speed of the fan accordingly. When the temperature is low, the fan rotates at a lower speed, and as the temperature increases, the fan speed also increases to provide better cooling. The system uses a driver circuit to supply sufficient current to the fan. This automatic operation helps in saving energy and maintaining comfortable environmental conditions. The Temperature Controlled Fan system is simple, cost-effective, and useful for homes, offices, computer systems, and industrial environments.

INTRODUCTION

- temperature controlled fan is an automatic cooling system that adjusts its speed based on the surrounding temperature. It is designed to provide efficient cooling while reducing unnecessary power consumption. Unlike traditional fans that operate at a constant speed, this system increases or decreases the fan speed depending on the temperature changes in the environment.
- This system is widely used in homes, electronic devices, and industrial applications to maintain an optimal temperature. It helps in improving energy efficiency, enhancing comfort, and protecting electronic components from overheating . Overall, a temperature controlled fan is a simple yet effective solution for smart cooling, combining sensing, processing, and control techniques in one system.

PROBLEM STATEMENT

In many places fans are operated manually. People switch the fan on or off depending on how they feel about the temperature. This manual operation has several disadvantages. Sometimes fans run even when cooling is not required, which wastes electricity. In computer systems and electronic devices, overheating can cause system failure. If cooling systems do not respond quickly to rising temperatures, components may get damaged. Manual control is not always reliable because humans cannot constantly monitor temperature. Therefore, there is a need for an automatic temperature-based fan control system. The system should detect the temperature and adjust the fan automatically. Such a system will reduce energy consumption, increase efficiency, and protect electronic equipment.

Limitations of Existing Methods

- **Manual Operation-** (Traditional fans require manual switching ON/OFF based on temperature.)
- **Energy Wastage**–(Fans may run continuously even when cooling is not required, which wastes electricity.)
- **No Automatic Temperature Detection**–(Many systems do not automatically sense temperature changes.)
- **Slow Response to Temperature Changes**–(Manual systems cannot respond quickly when temperature suddenly increases.)

PROPOSED DESIGN

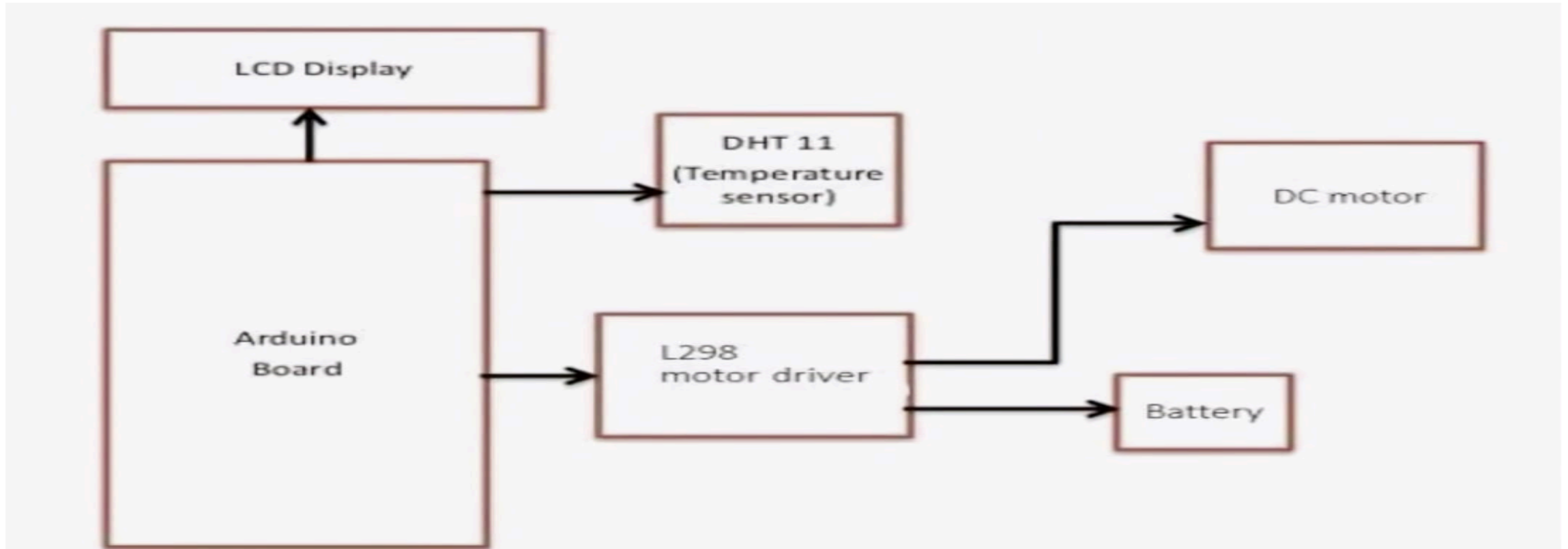


Fig.1.Block Diagram Temperature Controlled Fan

- **DHT11 Temperature Sensor** 🌡️ :-The DHT 11 Temperature Sensor measures the surrounding temperature . It converts temperature into an analog voltage signal . This signal is sent to the microcontroller.

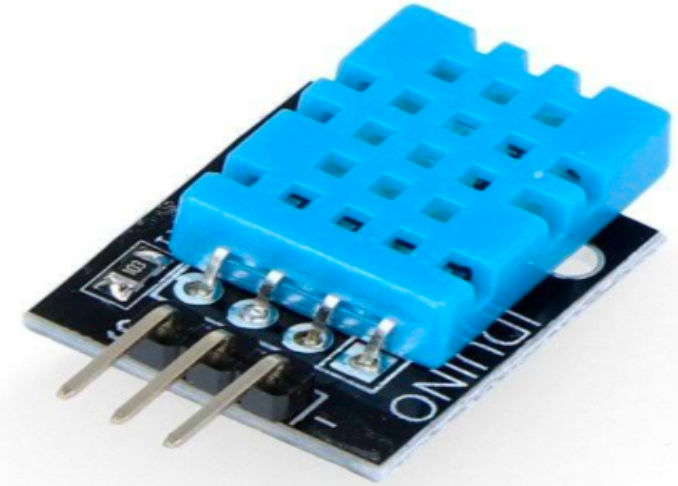


Figure.2: DHT 11 Temperature Sensor

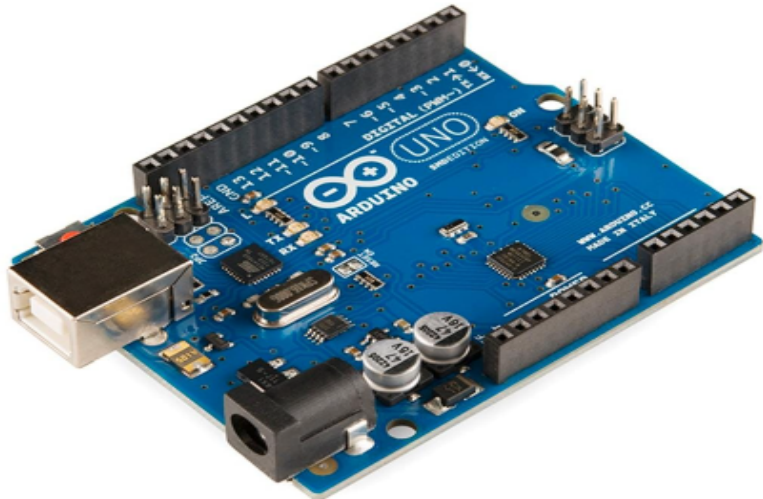


Figure.3: Arduino/UNO

- **ARDUINO**:-This is the Control Unit or the "brain" of the operation. It processes the data. It takes the analog signal from the DHT 11, converts it into a digital temperature value, and compares it against a pre-set threshold (e.g., "If Temp > 30°C, turn on fan").

- **LCD DISPLAY:-** LCD displays in temperature-controlled fans show real-time temperature readings and fan speed status.

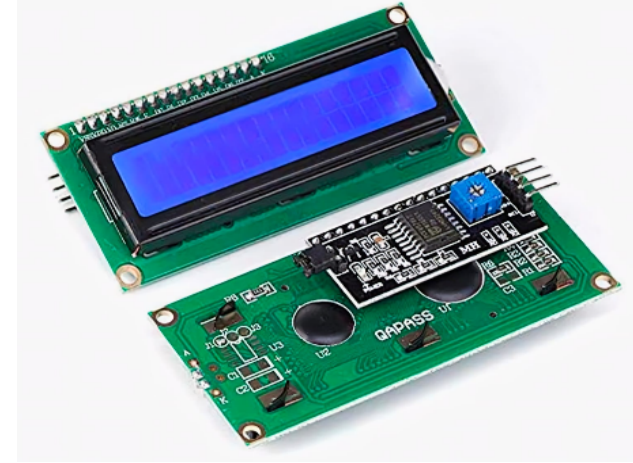


Figure.4: LCD Display



Figure.5: DC Fan

- **DC FAN:-** In temperature-controlled systems, they automatically adjust speed based on sensors like thermistors to optimize cooling, energy use, and noise. DC fans are direct current-powered cooling devices that use brushless motors for efficient airflow, commonly operating at voltage.

- **MOTOR DRIVER:-**Motor drivers like L298N (H-bridge for brushed DC) provide direction, speed control, and protection features such as overcurrent and thermal shutdown.

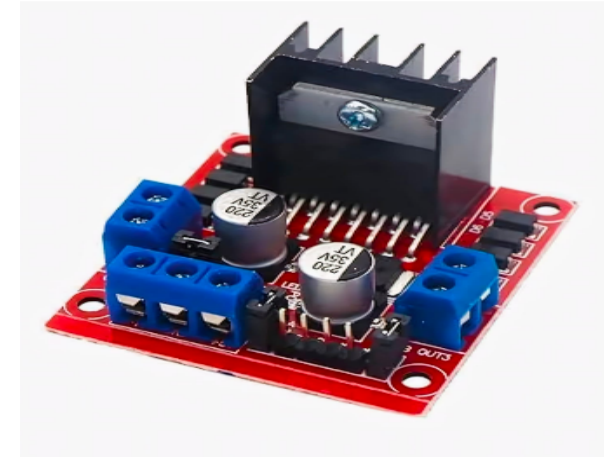


Figure.6: Motor Driver

The proposed design of the temperature controller fan consists of three main parts are temperature sensing unit, a control unit, and a fan driving unit. The temperature sensing unit uses a sensor such as LM35 or a thermistor. This sensor measures the surrounding temperature and converts it into an electrical signal. The control unit processes the signal received from the sensor. In a simple design, transistors or operational amplifiers may be used. In an advanced design, a microcontroller such as Arduino can be used for better control. The fan driving unit includes components like transistors, relays, or motor drivers that provide enough current to run the fan. The control signal from the controller activates this unit when required.

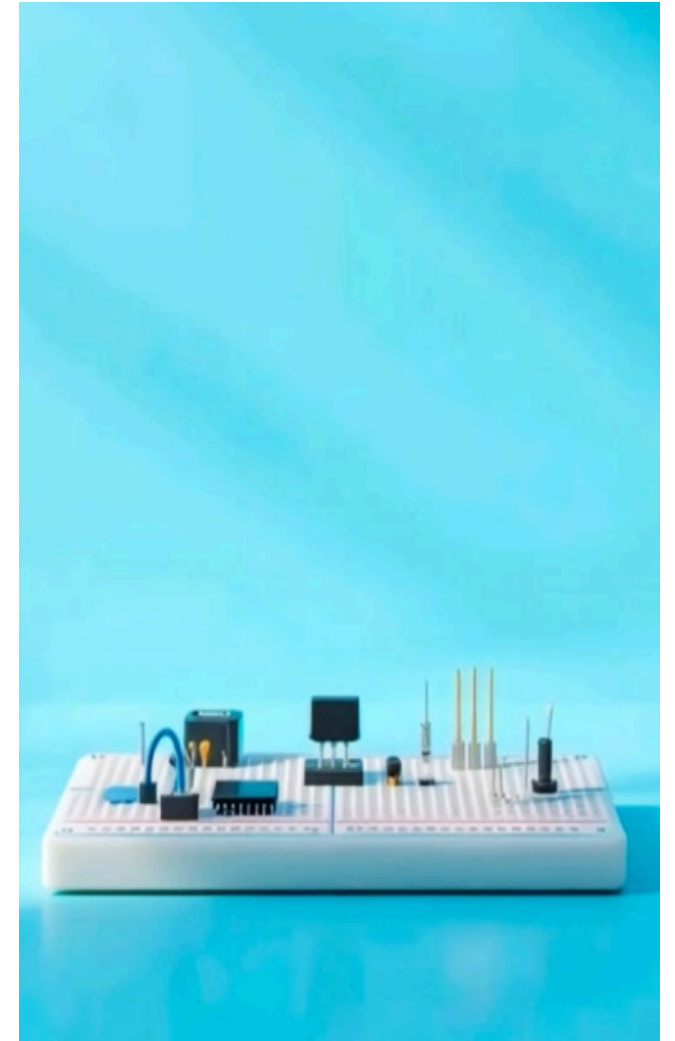
IMPLEMENTATION DETAILS

Hardware Integration

- ✓ DHT 11 sensor output connected to transistor base circuit.
- ✓ DC fan connected to collector Temperature-to-PWM duty cycle.
- ✓ Voltage Divider for base current control.
- ✓ Breadboard prototyping an PCB layout.
- ✓ Power supply regulation circuitry.

Software Development

- ✓ Arduino IDE for optional PWM control code.
- ✓ Analog read of DHT11 output via ADC pin.
- ✓ Temperature-to-PWM duty cycle mapping.
- ✓ Serial monitor for real-time temperature display.
- ✓ Threshold-based fan ON/OFF logic.
- ✓ Calibration routine for LM35 accuracy.

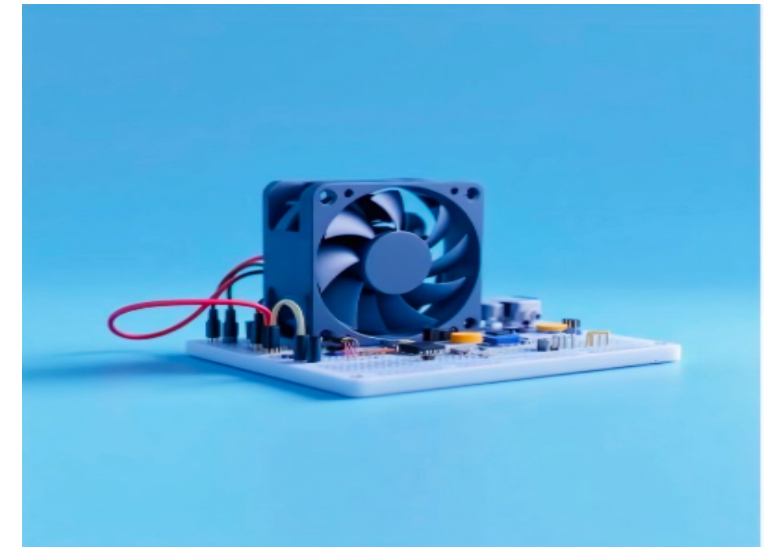


IMPLEMENTATION:

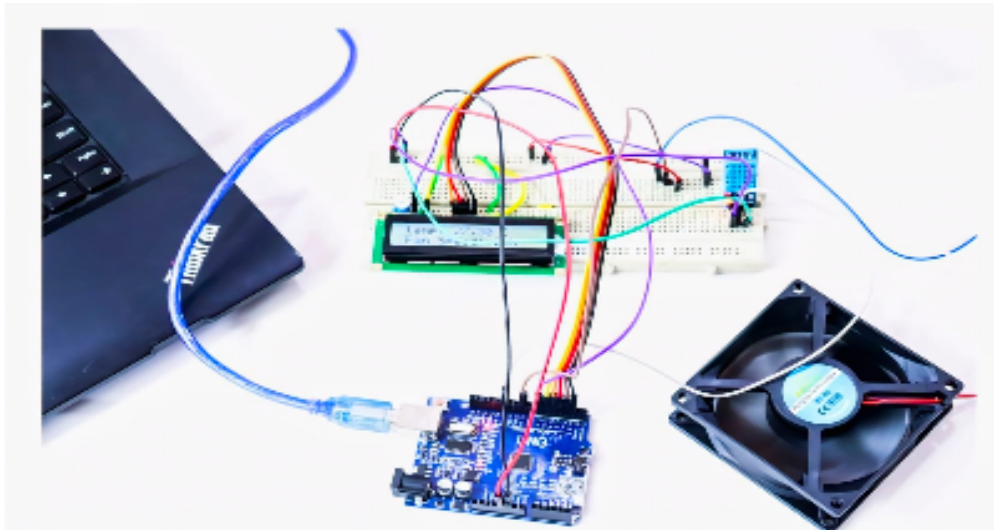
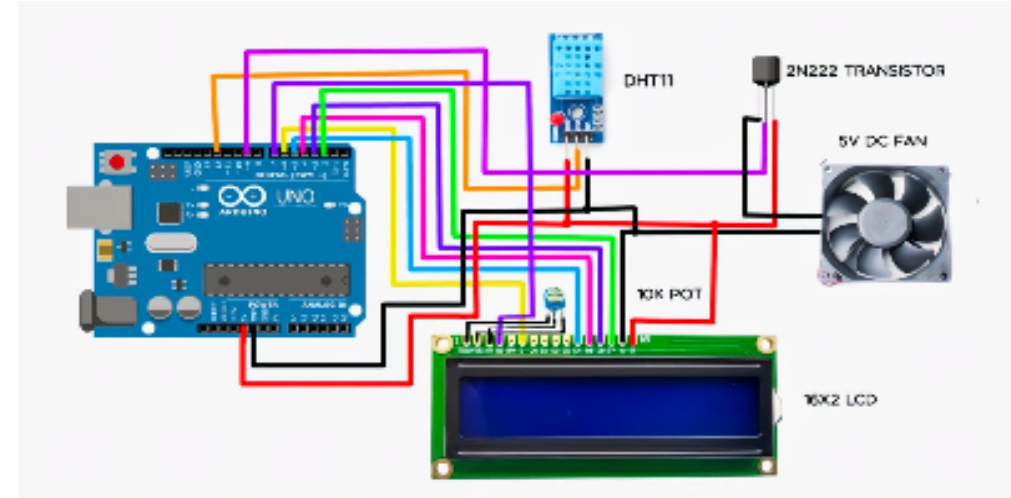
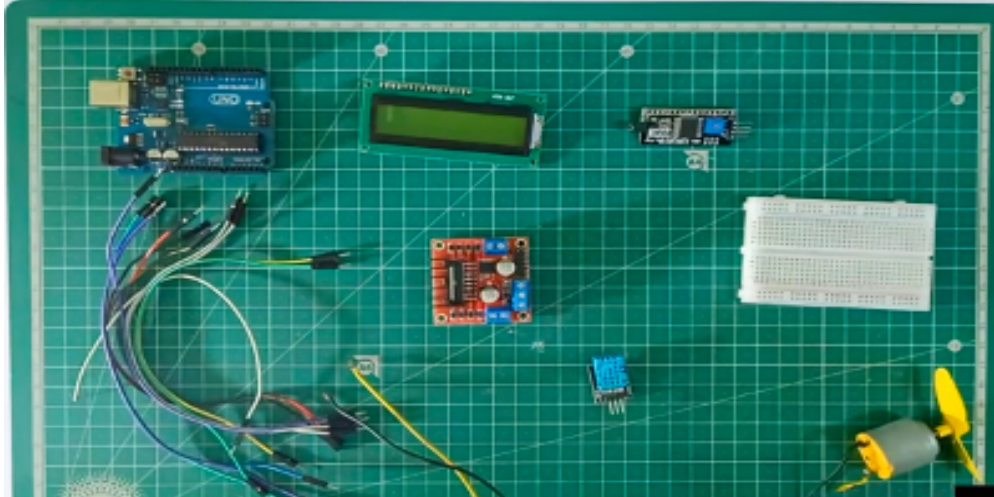
- Arduino Uno
- DHT 11Temperature Sensor
- DC Fan(5V/12V)
- Motor driver
- LCD display
- power Supply
- Breadboard
- Connecting wires

WORKING PRINCIPLE

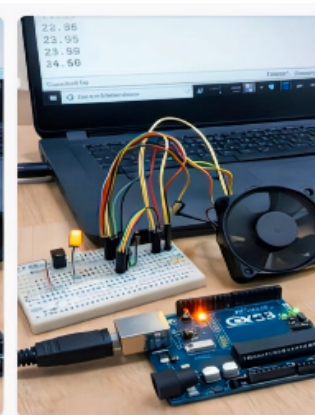
- The project was successfully implemented and tested. The DHT 11 sensor accurately measured the surrounding temperature and transmitted data to the Arduino microcontroller, Based on the temperature readings, the fan speed adjusted automatically.
- The temperature controlled fan system works effectively by automatically adjusting fan speed according to temperature variations. When the temperature rises, the fan speed increases, providing better cooling. When the temperature drops, the fan speed decreases, saving energy. This system improves comfort and reduces power consumption.
- Fan remained OFF when temperature was below 25°C.
- Fan started rotating when temperature reached 25°C.
- Fan speed increased as temperature increased above 30°C.



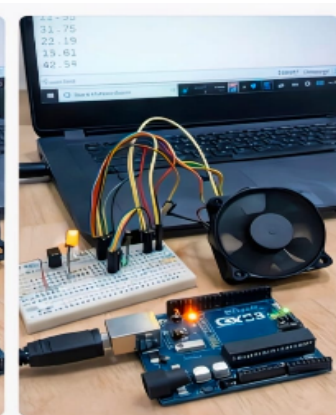
RESULTS AND DISCUSSIONS



Fan OFF when temperature < 25°C



Fan ON when temperature = 25°C



Fan speed increases when temperature > 30°C

ADVANTAGES :

1. Energy Efficient.
2. Automatic Operation.
3. Increased Comfort.
4. Extended Fan Life.
5. Cost Saving .
6. Smart control.
- 7.Environment friendly.

CONCLUSION AND FUTURE WORK

The temperature controller fan system is a simple and effective electronic project that demonstrates the concept of automatic temperature regulation. The project successfully uses a temperature sensor and control circuit to operate a fan based on environmental temperature. This system helps improve energy efficiency, protects electronic devices from overheating, and reduces manual effort. Due to its simplicity and low cost, it is suitable for educational demonstrations and practical learning in electronics and engineering courses. The temperature controlled fan using Arduino successfully automates the cooling system based on temperature changes. It helps in saving electricity and provides efficient cooling. In the future, the system can be improved by adding IoT technology for remote monitoring and control using mobile applications.

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THANK YOU